Lab 7: Fuzzing for Input Validation Bugs

Application: Car Rental Rate Technique: Hypothesis property-based testing

Objective

Generate unexpected text and None inputs for all processor functions, identify failures, and harden the application so malformed input never exposes parsing exceptions.

Figure 1 - Initial fuzz-test failures

```
$ .venv/bin/python -m pytest test_processor.py -q
FFF

FAILED test_sanitize_license_plate_no_crash
  s = None -> AttributeError: NoneType has no attribute strip

FAILED test_parse_trip_mileage_safe
  s = ''   -> ValueError: could not convert string to float
  s = None -> AttributeError: NoneType has no attribute split

FAILED test_format_rental_invoice_safe
  s = ':'  -> ValueError: invalid literal for int
  s = ''   -> IndexError: list index out of range
  s = None -> AttributeError: NoneType has no attribute split

3 failed
```

Hypothesis minimized each failure to a small, reproducible input.

Fuzz-test design

Each test uses @given(st.one_of(st.text(), st.none())) and verifies that its function returns the expected result type without throwing an exception.

Defensive fixes

1. sanitize_license_plate returns an empty string for non-text input.
2. parse_trip_mileage ignores blank, invalid, NaN, and infinite numeric fields.
3. format_rental_invoice validates the complete record and positive integral rental days.

Lab 7: Fixed Application Evidence

Figure 2 - Fixed processor run

```
==== Running sanitize_license_plate ====
Enter license plate with delimiters: Sanitized Plate: KA01AB1234

==== Running parse_trip_mileage ====
Parsed Mileage List: [100.5, 45.2, 80.0]

==== Running format_rental_invoice ====
Rental Invoice: Driver Rohan booked SUV for 3 days: Total $150.00
```

Figure 3 - Fuzz tests after the fix

```
$ .venv/bin/python -m pytest test_processor.py -q
...                                                                      [100%]
3 passed in 0.17s
```

Conclusion

All three completed Hypothesis fuzz tests now pass. The processor safely handles generated Unicode text and None input while retaining the expected valid rental workflow.

Commands used

```
python -m venv .venv
.venv/bin/pip install hypothesis pytest
.venv/bin/python processor.py
.venv/bin/python -m pytest test_processor.py -q
```

Lab 6 - Code Coverage Terminal Screenshot

Command: .venv/bin/python -m coverage report -m

```
rohan-sai@rohan-sai-Legion-Pro-5-16IRX8:~/Documents/SEM-5/se_jackfruit/Se-Lab6/L6_L7_02_car_rental_rate/Lab
6$ .venv/bin/python -m coverage report -m

Name                                              Stmts   Miss  Cover   Missing
-------------------------------------------------------------------------------

CodeCoverageTest_02_car_rental_rate.py               18      0   100%

car_rental_rate.py                                   51      0   100%

-------------------------------------------------------------------------------

TOTAL                                                69      0   100%


rohan-sai@rohan-sai-Legion-Pro-5-16IRX8:~/Documents/SEM-5/se_jackfruit/Se-Lab6/L6_L7_02_car_rental_rate/Lab
6$
```

Result: 100 percent line/code coverage.

Lab 6 - Branch Coverage Terminal Screenshot

Command: .venv/bin/python -m coverage report -m after branch run

```
rohan-sai@rohan-sai-Legion-Pro-5-16IRX8:~/Documents/SEM-5/se_jackfruit/Se-Lab6/L6_L7_02_car_rental_rate/Lab
6$ .venv/bin/python -m coverage report -m

Name                                              Stmts   Miss Branch BrPart  Cover   Missing
---------------------------------------------------------------------------------------------

BranchCoverageTest_02_car_rental_rate.py             33      0      0      0   100%

car_rental_rate.py                                   51      0     36      0   100%

---------------------------------------------------------------------------------------------

TOTAL                                                84      0     36      0   100%


rohan-sai@rohan-sai-Legion-Pro-5-16IRX8:~/Documents/SEM-5/se_jackfruit/Se-Lab6/L6_L7_02_car_rental_rate/Lab
6$
```

Result: 100 percent branch coverage.

Lab 7 - Fixed Application Terminal Screenshot

Command: .venv/bin/python processor.py

```
rohan-sai@rohan-sai-Legion-Pro-5-16IRX8:~/Documents/SEM-5/se_jackfruit/Se-Lab6/L6_L7_02_car_rental_rate/Lab
7$ .venv/bin/python processor.py
==== Running sanitize_license_plate ====
Enter license plate with delimiters - _ #: KA-01-AB_1234
Sanitized Plate: KA01AB1234

==== Running parse_trip_mileage ====
Enter CSV of mileages e.g. 100.5, 45.2, 80.0: 100.5, 45.2, invalid, , 80.0
Parsed Mileage List: [100.5, 45.2, 80.0]

==== Running format_rental_invoice ====
Enter rental record Driver:Days:Category: Rohan:3:SUV
Rental Invoice: Driver Rohan booked SUV for 3 days: Total $150.00

rohan-sai@rohan-sai-Legion-Pro-5-16IRX8:~/Documents/SEM-5/se_jackfruit/Se-Lab6/L6_L7_02_car_rental_rate/Lab
7$
```

Result: input sanitization, safe mileage parsing, and invoice generation complete.

Lab 7 - Hypothesis Fuzz-Test Terminal Screenshot

Command: .venv/bin/python -m pytest test_processor.py -q

```
rohan-sai@rohan-sai-Legion-Pro-5-16IRX8:~/Documents/SEM-5/se_jackfruit/Se-Lab6/L6_L7_02_car_rental_rate/Lab
7$ .venv/bin/python -m pytest test_processor.py -q

...

3 passed in 0.18s




rohan-sai@rohan-sai-Legion-Pro-5-16IRX8:~/Documents/SEM-5/se_jackfruit/Se-Lab6/L6_L7_02_car_rental_rate/Lab
7$
```

Result: all three Hypothesis fuzz tests pass.